# Apply Convolution as a NEW MODULE



Important

Since Convolution is a DOT product (between filter values and corresponding image va
It can be applied as if it is a NN Layer
**Input**: image values, **Weights** : filter values, **Activation**: Linear, **Bias**:0

***Remember**: for training, it is required to have the gradient of Output w.r.t both Input and weights*

- $\theta_{ij}$ ⇒ Filter weights
- **r, c** ⇒ row, column
- $a_{rc}$ ⇒ Activation value = Σ
- **θ** ⇒ Vevtor of all weights

- Activation function

$$a_{r,c} = \sum_{i=-a}^{a} \sum_{j=-b}^{b} x_{r+i,c+j} \cdot \theta_{ij}$$

- Essentially a dot product, similar to linear layer

$$a_{r,c} \sim X_{region}^{T} \cdot \theta$$

- Gradient w.r.t. the parameters

$$\frac{\partial a_{r,c}}{\partial \theta_{i,j}} = \sum_{r=0}^{N-2a} \sum_{c=0}^{N-2b} x_{r-i,c-j}$$

**Weights**: filter values

**Input**: pixel values